

Amit Kumar Mohanty

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Target Role: Junior GenAI Developer | Miratech

PROFESSIONAL SUMMARY

GenAI Engineer with 1+ year of hands-on experience building production-grade Generative AI systems, Retrieval-Augmented Generation (RAG) pipelines, and Python backend services. Skilled in developing LLM-powered APIs with FastAPI, agent-based workflows with tool/function calling, and vector search using FAISS and Chroma. Experienced with LangChain, LangGraph, and OpenAI/Anthropic LLM APIs, and in debugging hallucinations and retrieval failures using faithfulness and relevance evaluation. Focused on writing clean, testable, maintainable code and moving GenAI POCs into reliable, scalable production on AWS.

EDUCATION

K.R. Mangalam University, Gurgaon

July 2021 - July 2025

B.Tech in Computer Science and Engineering (Specialization in AI/ML)

Birla Institute of Technology and Science, Pilani (BITS Pilani)

July 2026 - July 2028

M.Tech in Artificial Intelligence and Machine Learning (Pursuing)

EXPERIENCE

Research Associate - AI, Keywords Studios India - Gurgaon, Haryana

May 2025 - Present

- Built and optimized agent-based GenAI workflows with tool/function calling that autonomously execute complex multi-step tasks, improving task-completion reliability.
- Developed Python data-ingestion pipelines (ingestion, chunking, embedding) and delivered 85% of self-generated training datasets to production for LLM readiness.
- Debugged retrieval and quality issues across LLM outputs, collaborating with senior engineers to move POCs into production; awarded "Rookie of the Year 2025" for exceeding all KPIs.

Artificial Intelligence Intern, Infosys - Remote

Nov 2024 - Feb 2025

- Engineered a Python LLM service using LangChain and LangSmith, boosting response accuracy by 35% and integrating Large Language Models into backend workflows.
- Reduced hallucinations by 20% through prompt engineering and automated RAG-style evaluation of faithfulness and relevance.
- Deployed LLM features that cut system latency by 25%, collaborating with a 12-engineer team using Git and standard software development practices.

PROJECTS

Dynamic Screen Companion (RAG + LLM Backend)

June 2025 - Present

- Built a real-time GenAI assistant integrating LLM APIs (Gemini/OpenAI) with a retrieval layer over screen context to generate grounded, low-hallucination responses.
- Engineered a Python FastAPI backend exposing REST APIs for concurrent LLM inference streams, containerized with Docker and CI/CD for scalable deployment.
- Implemented an embedding + vector-search pipeline (chunking, embeddings, retrieval) maintaining 95%+ extraction accuracy across varied inputs.

Diabetic Optiscan (Peer-reviewed research)

Jan 2025 - June 2025

- Built a Vision Transformer (ViT-B/16) system on the APTOS dataset to detect diabetic retinopathy; co-authored a peer-reviewed paper on the design and accuracy gains.
- Designed a custom wavelet technique that boosted detection precision by 50% over legacy methods.

SKILLS

Languages & Backend: Python, REST API design, Backend Service Development, FastAPI, SQL, C++, Git

GenAI & RAG: LLMs, RAG (ingestion, chunking, embedding, retrieval, response generation), LangChain, LangGraph, OpenAI API, Anthropic (Claude) API, Agent Workflows, Tool/Function Calling, Prompt Engineering, Tokens & Context Windows, Hallucination Debugging, RAG Evaluation (Faithfulness, Relevance)

Vector Search & Data: Embeddings, Vector Databases (FAISS, Chroma, Pinecone), Hybrid Search, Data Pipelines, Pandas, NumPy, MongoDB, MySQL

Cloud & MLOps: AWS, GCP (Vertex AI), Docker, Kubernetes, CI/CD, MLflow, Model Serving & Monitoring

ML Frameworks: PyTorch, TensorFlow, Hugging Face, Transformers, Scikit-learn, OpenCV

CERTIFICATIONS

- Ducat India - Data Science Professional Training (Python, ML, statistics, predictive modeling), Apr 2024 - Apr 2025.
- ISRO - AI/ML for Geodata Analysis (applied ML/DL to remote-sensing imagery).
- Actively pursuing AWS and GenAI/LLM engineering certifications.